

```

snavya@snavya1955: ~/os
snavya@snavya1955:~$ cs ~/os
Command 'cs' not found, but can be installed with:
sudo apt install csound
snavya@snavya1955:~$ cd ~/os
snavya@snavya1955:~/os$ ls
1          bin          file1.txt      memory1.c      os1.1.7.c     os5.c         program7.c     signal1       signal7.c
1.1.1c    copy_low.txt    filecopy      mermory.c      os1.1.8.c     os6.c         project        signal1.c    source.txt
1.c       copy_standard.txt filecopy1.c   mermory.c      os1.1.c       os7.c         quotes        signal2.c    src
2         cow          filecopy1     new           os1.2.c       os8.c         quotes.sh     signal2.c    std
2.c       cow.c         filecopy1.c   op.save       os1.3.c       output.txt    quotes.sh     signal3.c    std.c
3         destination.txt filecopy2     os.c          os1.4.c       parser        redirect      signal3.c    std.c.save
3.c       destination2.txt filecopy2.c   os1.1.1.c     os1.c         parser.c      redirect.c    signal4.c    test.txt
4.c       destination_std.txt filecopy_std os1.1.2.c     os1.xc        parser1.c     sample.txt    signal4.c    test1.txt
'Another File.txt' docs      filecopy_std.c os1.1.3.c     os14.c        parser1.c     shell         signal5.c    test2.txt
FileDescriptors escape.c   input.txt      os1.1.4.c.save os15.c        process       shell.c       signal5.c    tokenizer.c
LinuxLab  escape.c   memory        os1.1.5.c     os2.c         process.c     shell_history signal6.c
'My File.txt' exec      memory.c      os1.1.6.c     os3.c         program1.c    shell_history.c.save signal6.c
MyProject fifo_project memory1       os1.1.6.c     os4.c         program7      shell_history.c.save signal7

snavya@snavya1955:~/os$ pwd
/home/snavya/os
snavya@snavya1955:~/os$ ls -i source.txt
11337 source.txt
snavya@snavya1955:~/os$ stat source.txt
  File: source.txt
  Size: 70          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 11337       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/  snavya)   Gid: ( 1000/  snavya)
Access: 2026-10-07 02:45:28.222334968 +0000
Modify: 2026-08-05 03:30:25.109069490 +0000
Change: 2026-08-05 03:30:25.109069490 +0000
 Birth: 2026-08-05 03:30:25.106357270 +0000
snavya@snavya1955:~/os$ ls -i source.txt
11337 source.txt
snavya@snavya1955:~/os$ ln source.txt hardlink.txt
snavya@snavya1955:~/os$ ls -li source.txt hardlink.txt
11337 -rw-r--r-- 2 snavya snavya 70 Aug  5 03:30 hardlink.txt
11337 -rw-r--r-- 2 snavya snavya 70 Aug  5 03:30 source.txt
snavya@snavya1955:~/os$ ln source.txt hardlink.txt
ln: failed to create hard link 'hardlink.txt': File exists
snavya@snavya1955:~/os$ ls -li source.txt hardlink.txt
11337 -rw-r--r-- 2 snavya snavya 70 Aug  5 03:30 hardlink.txt

```

```

sravya@Sravya1955:~/os$ ls -li source.txt hardlink.txt symlink.txt
11337 -rw-r--r-- 2 sravya sravya 70 Aug  5 03:30 hardlink.txt
11337 -rw-r--r-- 2 sravya sravya 70 Aug  5 03:30 source.txt
45171 lrwxrwxrwx 1 sravya sravya 10 Oct  7 03:12 symlink.txt -> source.txt
sravya@Sravya1955:~/os$ ln -s source.txt symlink.txt
ln: failed to create symbolic link 'symlink.txt': File exists
sravya@Sravya1955:~/os$ ls -li source.txt hardlink.txt symlink.txt
11337 -rw-r--r-- 2 sravya sravya 70 Aug  5 03:30 hardlink.txt
11337 -rw-r--r-- 2 sravya sravya 70 Aug  5 03:30 source.txt
45171 lrwxrwxrwx 1 sravya sravya 10 Oct  7 03:12 symlink.txt -> source.txt
sravya@Sravya1955:~/os$ find . -inum 11337
./hardlink.txt
./source.txt
sravya@Sravya1955:~/os$ find . -inum 11337
./hardlink.txt
./source.txt
sravya@Sravya1955:~/os$ rm source.txt
sravya@Sravya1955:~/os$ ls -li hardlink.txt symlink.txt
11337 -rw-r--r-- 1 sravya sravya 70 Aug  5 03:30 hardlink.txt
45171 lrwxrwxrwx 1 sravya sravya 10 Oct  7 03:12 symlink.txt -> source.txt
sravya@Sravya1955:~/os$ cat hardlink.txt
Hello!
This is Operating Systems Lab.
Learning system calls in Linux.
sravya@Sravya1955:~/os$ rm source.txt
rm: cannot remove 'source.txt': No such file or directory
sravya@Sravya1955:~/os$ ls -li hardlink.txt symlink.txt
11337 -rw-r--r-- 1 sravya sravya 70 Aug  5 03:30 hardlink.txt
45171 lrwxrwxrwx 1 sravya sravya 10 Oct  7 03:12 symlink.txt -> source.txt
sravya@Sravya1955:~/os$ cat hardlink.txt
Hello!
This is Operating Systems Lab.
Learning system calls in Linux.
sravya@Sravya1955:~/os$ cat symlink.txt
cat: symlink.txt: No such file or directory
sravya@Sravya1955:~/os$ pwd
/home/sravya/os
sravya@Sravya1955:~/os$ nano mmap_file.c
sravya@Sravya1955:~/os$ nano mmap.txt

```

GNU nano 7.2

```
#include <stdio.h>
#include <fcntl.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd;
    struct stat st;
    char *mapped;

    fd = open("mmap.txt", O_RDWR);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        perror("fstat");
        close(fd);
        return 1;
    }

    mapped = mmap(NULL, st.st_size,
                  PROT_READ | PROT_WRITE,
                  MAP_SHARED, fd, 0);

    if (mapped == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }
}
```

```

        close(fd);
        return 1;
    }

    printf("Original content:\n");
    write(STDOUT_FILENO, mapped, st.st_size);

    mapped[0] = 'H';

    munmap(mapped, st.st_size);
    close(fd);

    printf("\nFile modified successfully.\n");

    return 0;
}

```

```

sravya@Sravya1955:~/os$ nano mmap_file.c
sravya@Sravya1955:~/os$ gcc mmap_file.c -o mmap_file
sravya@Sravya1955:~/os$ ./mmap_file
Original content:
Hello!
This is memory mapped file I/O.
Operating Systems Lab.

File modified successfully.
sravya@Sravya1955:~/os$ cat mmap.txt
Hello!
This is memory mapped file I/O.
Operating Systems Lab.

sravya@Sravya1955:~/os$ cat mmap.txt
Hello!
This is memory mapped file I/O.
Operating Systems Lab.

sravya@Sravya1955:~/os$ nano mmap_file.c
sravya@Sravya1955:~/os$ |

```